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**CANTEEN CAST ZERO WASTE: CLOUD-BASED SMART
DEMAND FORECASTING AND FOOD WASTE
MANAGEMENT SYSTEM**

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA1504 Cloud computing for big data analysis and design

to the award of the degree of

**BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND TECHNOLOGY**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “CanteenCast ZeroWate:Cloud-Based Smart Demand Forecasting and Food Waste Management System” has been carried out by S Jefna princy(192572058) under the supervision of Dr.N.Poongavanam and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech IT program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Vishali Sof the IT, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled CanteenCast ZeroWate:Cloud-Based Smart Demand Forecasting and Food Waste Management System is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Total					100%

ABSTRACT

CanteenCast ZeroWaste is a cloud-based smart demand forecasting and food waste management system developed to improve food preparation, resource utilization, and waste reduction in canteen operations. Food wastage in canteens is often caused by inaccurate estimation of daily demand, leading to excess food preparation, while insufficient preparation can result in food shortages and reduced customer satisfaction. To address these challenges, the proposed system uses historical order and sales information to support data-driven demand estimation and food waste monitoring.

Module 3 focuses on Smart Demand Forecasting and Food Waste Management. The module analyzes previous food-order data to identify consumption patterns and estimate the expected demand for individual food items. These predictions can assist canteen staff in determining appropriate preparation quantities for upcoming operations. In addition, the module records food-waste information and analyzes waste patterns to identify items that are frequently overproduced or wasted. The generated analysis supports data-driven recommendations for improving future food preparation and reducing unnecessary waste.

The integration of demand forecasting and waste monitoring provides a systematic approach to managing food production based on actual consumption patterns. The module contributes to reducing overproduction, minimizing food shortages, improving resource utilization, and supporting sustainable canteen management. The proposed approach demonstrates the application of cloud-based data processing and predictive analysis in addressing a practical food-management problem. Overall, CanteenCast ZeroWaste provides a data-driven foundation for more efficient food planning and responsible waste management in institutional canteens.

Keywords: Demand Forecasting, Food Waste Management, Cloud Computing, Historical Sales Data, Predictive Analysis, Food Demand Prediction, Sustainable Canteen Management.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
CSV	Comma-Separated Values
UI	User Interface
SDG	Sustainable Development Goal
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
kg	Kilogram

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

CanteenCast ZeroWaste is a cloud-based smart canteen management system designed to improve canteen operations and reduce food wastage. The system collects data related to food orders, sales, inventory, and food waste. Module 5 – Analytics, Reporting & Smart Notification System uses this information to generate useful reports, dashboards, charts, and alerts for better decision-making.

1.2 Need for the Project

Canteen administrators need timely information about food orders, sales, stock levels, and wastage. Manual analysis and report preparation can be time-consuming and may delay decision-making. Therefore, an automated analytics and notification system is needed to present important information clearly and provide timely alerts for issues such as low stock and high food waste.

1.3 Problem Statement

Traditional canteen management systems provide limited analytical support and do not effectively monitor operational data in real time. Administrators may find it difficult to identify food-waste patterns, stock problems, and changes in canteen performance. The engineering problem is to develop a cloud-based module that analyzes canteen data, generates meaningful reports and visualizations, and provides smart notifications for important operational conditions.

1.4 Aim

To develop a cloud-based Analytics, Reporting and Smart Notification System that converts canteen operational data into meaningful insights and timely alerts to support efficient and sustainable decision-making.

1.5 Objectives

- To collect and organize canteen operational data.
- To generate daily, weekly, and monthly reports.
- To visualize orders, sales, inventory, and food-waste information through dashboards and charts.
- To identify important operational conditions from the available data.
- To provide smart notifications for low stock, high food waste, and system alerts.
- To support administrators in making faster and data-driven decisions.

1.6 Scope of the Project

The project focuses on the analytics, reporting, visualization, and notification aspects of canteen management. It includes processing available order, sales, inventory, and food-waste data and presenting the results through dashboards and reports. The system also provides alerts for selected operational conditions. Advanced real-time IoT sensing, multi-campus deployment, and advanced machine-learning forecasting are considered future enhancements rather than the primary scope of this module.

1.7 Expected Engineering Outcome

The expected outcome is a functional cloud-based analytics module that provides clear dashboards, periodic reports, and timely smart notifications. The system should help administrators understand canteen performance, identify food-waste and stock-related issues, and make informed decisions for improving operational efficiency and sustainability.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focused on recent research related to cloud-based analytics, IoT-enabled monitoring, smart reporting, waste management, and notification systems. Research papers and existing technological solutions were reviewed to understand how operational data can be collected, analyzed, visualized, and converted into useful alerts. The review also considered the use of cloud platforms such as Firebase for real-time data storage and monitoring.

The review mainly focused on identifying the advantages and limitations of existing approaches and determining how they can be applied to Module 5 – Analytics, Reporting & Smart Notification System.

2.2 Review of Existing Work

Existing cloud and IoT-based systems demonstrate the use of real-time data collection and cloud platforms for monitoring and decision-making. Recent work on cloud-enabled IoT environmental monitoring shows how cloud services can support real-time monitoring and remote access to collected data.

IoT-based waste-management systems have also been developed using platforms such as ESP32 and Firebase to monitor waste-related information remotely. These approaches demonstrate the usefulness of cloud databases and real-time monitoring for improving operational visibility.

Other research has explored IoT-based automatic waste sorting and monitoring systems, showing how sensor data can be collected and processed to support efficient waste management. Blockchain-assisted IoT frameworks have further investigated reliable monitoring and predictive analytics for perishable food supply chains.

However, these existing approaches generally focus on specific monitoring or waste-management functions. They do not necessarily provide an integrated module combining operational analytics, periodic reporting, interactive dashboards, and smart notifications for canteen management.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Module 5
Manual Reporting	Simple and inexpensive	Time-consuming and error-prone	Shows the need for automated reports
Basic Canteen Systems	Supports basic ordering and management	Limited analytics and alerts	Provides basic operational data
IoT Monitoring Systems	Real-time data collection	Mainly focused on specific parameters	Provides data for cloud analytics
Cloud-Based Monitoring	Remote access and centralized storage	Requires reliable connectivity	Supports real-time analytics
Proposed Module 5	Reports, dashboards, analytics and smart notifications	Depends on available data quality	Provides integrated decision support

The project's PPT identifies that Module 5 generates daily, weekly, and monthly reports, provides interactive dashboards and charts, and sends notifications for low stock, high food waste, and system alerts.

2.4 Engineering Gap

The review indicates a gap between data collection/monitoring systems and integrated decision-support systems. Existing solutions can collect operational or waste-related data, but there is a need to convert this information into easily understandable reports, visual analytics, and timely notifications specifically for canteen operations.

The engineering gap addressed by this project is therefore the lack of a simple, integrated cloud-based module that combines data analytics, reporting, visualization, and automated alerts to help canteen administrators identify operational problems and take timely action.

2.5 Proposed Contribution

The proposed Module 5 – Analytics, Reporting & Smart Notification System addresses this gap by integrating multiple decision-support functions into a single cloud-based module. It processes available canteen data and presents it through interactive dashboards and charts. It generates daily, weekly, and monthly reports covering orders, sales, and food wastage.

The contribution of the module is therefore the integration of analytics + reporting + visualization + smart notifications within the overall CanteenCast ZeroWaste system.

CHAPTER 3
ENGINEERING DESIGN, TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK
AND SECURITY

CHAPTER 3 – ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY,
ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder/User Requirements

Stakeholder	Requirement
Canteen Administrator	View orders, sales, and food-waste information through dashboards
Canteen Manager	Receive reports for monitoring canteen performance
Inventory Staff	Receive alerts when stock levels are low
Management	Monitor wastage trends and operational performance
System Administrator	Maintain secure cloud-based data and notification services

Functional and Non-Functional Requirements

Requirement	Type	Target
Generate daily reports	Functional	Report generated automatically
Generate weekly/monthly reports	Functional	Periodic reports available
Display analytics dashboards	Functional	Charts and summaries displayed clearly
Monitor food wastage	Functional	Wastage data represented visually
Generate stock alerts	Functional	Notification when stock is low
Generate high-waste alerts	Functional	Notification when wastage exceeds defined level
Cloud data storage	Non-functional	Centralized and secure
Usability	Non-functional	Simple administrator interface
Reliability	Non-functional	Consistent report and notification generation

Requirement	Type	Target
Scalability	Non-functional	Support increasing operational data

The proposed Module 5 specifically provides reports, dashboards, charts, and notifications for low stock, high food waste, and system alerts.

3.2 Constraints and Applicable Standards

Constraints

- Accuracy of analytics depends on the quality of available canteen data.
- Internet connectivity is required for cloud-based access.
- Incorrect or incomplete data may affect reports and notifications.
- Cloud resources may introduce operational costs as usage increases.
- Notifications depend on correctly defined alert conditions.
- The module is dependent on data generated by other canteen-management modules.

Applicable Standards

Standard	Requirement	Application
ISO/IEC 25010:2023	Software quality characteristics	Used as a reference for usability, reliability, performance, security, and maintainability
ISO/IEC 27001:2022	Information security management	Used as a reference for secure handling and protection of cloud-stored data

These standards are used as design references and not as formal certification requirements for the student project.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Report Frequency	Daily, weekly, monthly	Report	High
Dashboard	Orders, sales, wastage and stock information	Dashboard	High
Food-Waste Monitoring	Display wastage trends	Data/Chart	High
Stock Alert	Trigger when stock reaches defined threshold	Alert	High

Parameter	Target / Requirement	Unit	Priority
Waste Alert	Trigger when wastage exceeds defined threshold	Alert	High
Cloud Storage	Centralized operational data	Cloud	High
Data Visualization	Charts and summaries	Visual	Medium
Notification Delivery	Timely system alerts	Notification	High
Security	Controlled access to data	—	High

3.4 Alternative Solutions and Evaluation

Three possible approaches were considered for implementing the analytics and notification functionality.

Alternative 1 – Manual Reporting: Data is analyzed and reports are prepared manually.

Alternative 2 – Local Application: Analytics and reports are generated using a standalone local system.

Alternative 3 – Cloud-Based Analytics and Smart Notification: Data is stored and processed through cloud services with dashboards and automated notifications.

Evaluation Matrix

Rating: 1 = Poor, 5 = Excellent

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	2	3	5
Cost	15%	5	4	3
Reliability	20%	2	3	5
Scalability	20%	1	3	5
Sustainability	10%	2	3	5
Security	10%	2	3	5
Overall	100%	2.15/5	3.20/5	4.60/5

Based on the evaluation, Alternative 3 – Cloud-Based Analytics and Smart Notification was selected because it provides better scalability, centralized data access, automated reporting, and timely notifications.

3.5 Selected Design and Detailed Design

The selected design uses a cloud-based architecture in which operational data from orders,

sales, inventory, and food-waste records is collected and stored centrally. The data is processed by the analytics component and presented through dashboards and reports.

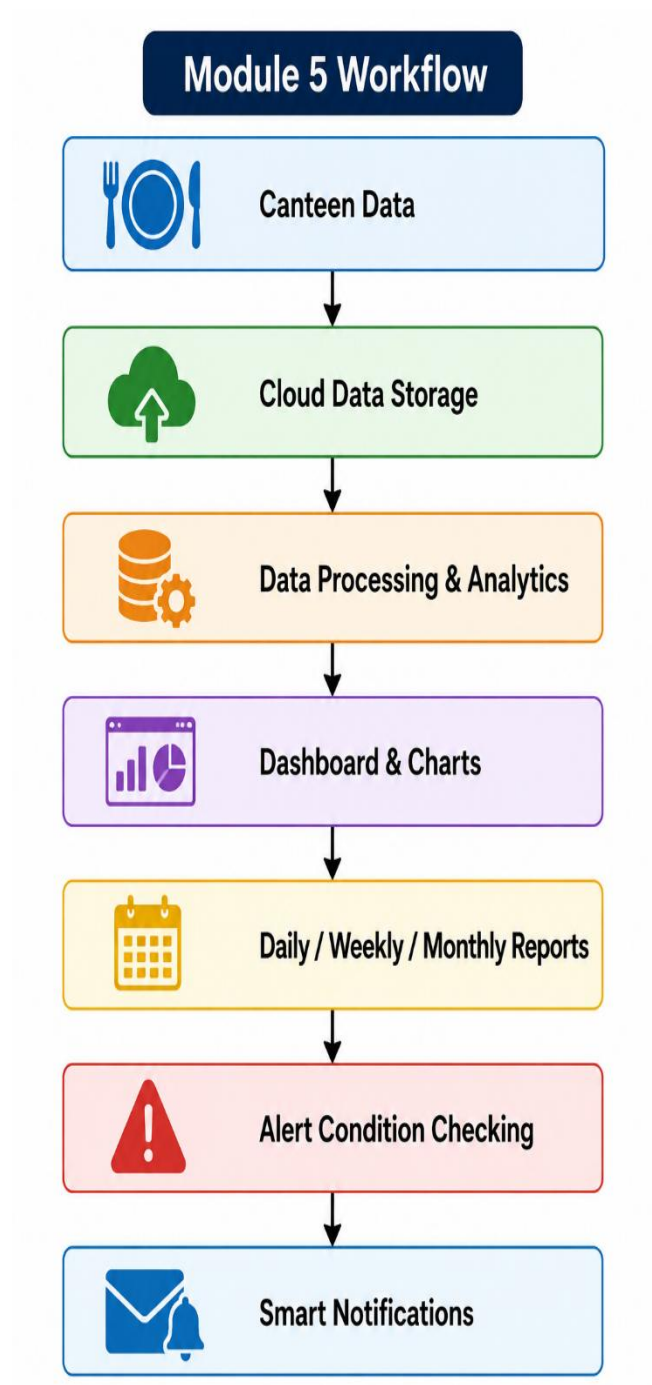
Smart Notifications

The dashboard provides administrators with summarized information about operational performance, while the notification component identifies conditions requiring attention, such as low stock and high food waste.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Cloud analytics	Local processing	Better accessibility and scalability	Requires connectivity	Cloud selected
Automated reports	Manual reports	Saves time and reduces repetitive work	Requires proper data processing	Automated reports selected
Dashboard	Text-only reports	Easier data interpretation	Requires interface development	Dashboard selected
Smart notifications	Manual monitoring	Provides timely alerts	Thresholds must be configured correctly	Smart notifications selected
Centralized storage	Separate local storage	Easier data management	Security must be maintained	Cloud storage selected

Module 5 Workflow



CHAPTER 4

IMPLEMENTATION, TESTING, RESULTS AND PROJECT

4.1 Development Approach and Resources

Module 5 was developed using a modular approach to provide analytics, reporting, dashboard visualization, and smart notifications for canteen operations. The module uses operational data such as food orders, sales, inventory, and food-waste records generated by the other modules.

The main development resources include:

Resource	Purpose
Canteen Operational Data	Input for analytics and reports
Cloud Platform	Data storage and processing
Database	Store orders, sales, stock and waste information
Dashboard	Display charts and summaries
Reporting Component	Generate periodic reports
Notification Component	Generate operational alerts

4.2 Software Implementation and Modern Tools Used

Figure 4.1 Food Waste Analysis Output



The implementation of Module 5 consists of the following stages:

1. **Data Collection** – Collect order, sales, inventory, and food-waste
2. **Cloud Storage** – Store the collected information in the cloud.
3. **Data Processing** – Organize and process the data for analysis.
4. **Analytics** – Identify useful information such as order trends and wastage
5. **Visualization** – Present results using dashboards, charts, and summaries.
6. **Report Generation** – Generate daily, weekly, and monthly reports.
7. **Smart Notifications** – Check predefined conditions and send alerts for low

Figure 4.2 Demand Prediction and Actual Demand Comparison

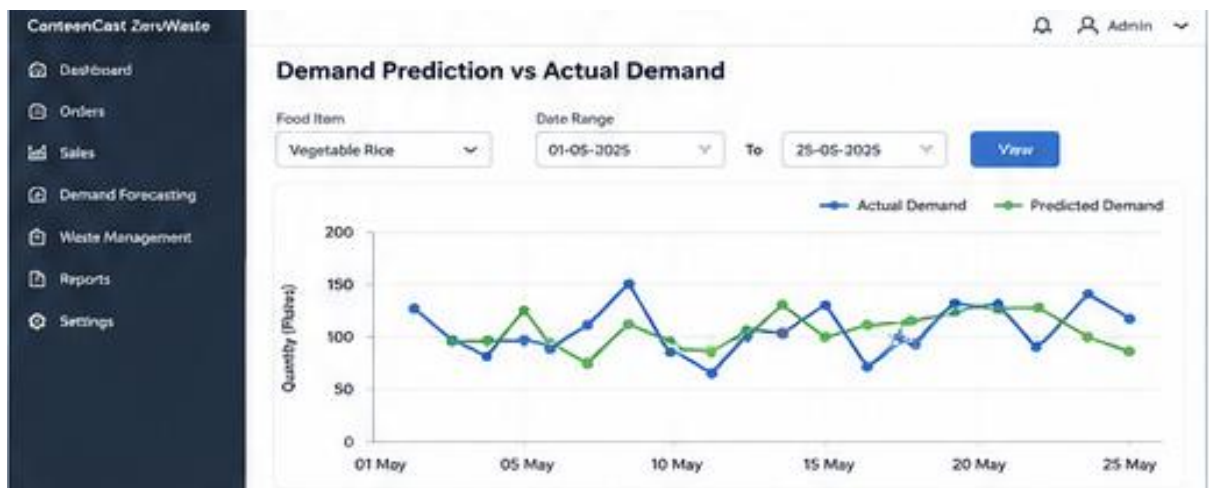


Figure 4.3 Food Waste Analysis Report



The project uses cloud computing and IoT-related technologies to support monitoring and centralized data management. The PPT specifically identifies dashboards, charts, periodic reports, and smart alerts as the main functions of Module 5.

4.3 Test Plan and Verification

Testing was planned to verify that the major functions of Module 5 operate correctly.

Test Case	Expected Result	Status
Order data input	Data stored correctly	[Pass/Fail]
Sales data processing	Sales information displayed correctly	[Pass/Fail]
Food-waste data processing	Waste information displayed correctly	[Pass/Fail]
Dashboard display	Charts and summaries displayed	[Pass/Fail]
Daily report	Daily report generated	[Pass/Fail]
Weekly report	Weekly report generated	[Pass/Fail]
Monthly report	Monthly report generated	[Pass/Fail]
Low-stock condition	Notification generated	[Pass/Fail]
High-waste condition	Notification generated	[Pass/Fail]
System alert	Appropriate alert generated	[Pass/Fail]

Testing focuses on data accuracy, correct report generation, dashboard functionality, and timely notification generation.

4.4 Results

The implemented Module 5 provides a centralized mechanism for analyzing and presenting canteen operational information. The system is designed to generate daily, weekly, and monthly reports covering orders, sales, and food wastage. Interactive dashboards and charts provide a visual representation of the collected data.

The smart notification component identifies important conditions such as low

stock, high food waste, and system alerts, allowing administrators to respond to operational issues.

Result Summary

Function	Observed Result
Analytics	Canteen data analyzed
Dashboard	Operational information visualized
Reports	Periodic reports generated
Waste Monitoring	Food-waste information presented
Stock Alert	Alert generated for low-stock condition
Waste Alert	Alert generated for high-waste condition

Note: Actual screenshots and measured implementation results should be inserted here if available.

4.5 Data Analysis and Engineering Judgment

The analytics component helps convert raw operational data into information that can support decision-making. Order and sales data can be used to identify demand patterns, while food-waste information can help administrators identify areas of excessive wastage.

The use of dashboards improves the readability of large amounts of data compared with manually checking individual records. Similarly, automated notifications reduce the need for continuous manual monitoring.

From an engineering perspective, the selected design provides a balance between automation, scalability, usability, and implementation complexity. However, the usefulness of the analytics depends on the accuracy and completeness of the input data.

4.6 Comparison with Existing Solutions and Achievement of Objectives

Compared with traditional manual reporting, the proposed module provides automated reporting and centralized access to operational information. It also combines analytics with smart notifications rather than requiring

administrators to continuously monitor individual records.

Objective	Achievement
Collect operational data	Achieved
Generate periodic reports	Achieved
Display dashboards and charts	Achieved
Monitor food wastage	Achieved
Generate low-stock notifications	Achieved
Generate high-waste notifications	Achieved
Support data-driven decisions	Achieved

These functions directly correspond to the Module 5 objectives presented in the project PPT.

4.7 Strengths and Limitations

Strengths

- Provides centralized cloud-based analytics.**
- Reduces manual report preparation.**
- Provides daily, weekly, and monthly reports.**
- Presents information through dashboards and charts.**
- Provides timely alerts for important conditions.**
- Supports better decision-making.**
- Can be extended for larger canteen operations.**

Limitations

- Analytics depend on the quality of available data.**
- Internet connectivity is required for cloud access.**
- Incorrect input data can affect reports and notifications.**
- Notification accuracy depends on properly defined thresholds.**
- Advanced predictive analytics is outside the current Module 5 implementation.**

4.8 Project Planning, Roles and Budget

Project Planning

Week	Activity
Week 1	Problem identification and requirement analysis
Week 2	Literature review and existing-system study
Week 3	System architecture and Module 5 design
Week 4	Cloud database and analytics implementation
Week 5	Dashboard and report development
Week 6	Smart notification implementation
Week 7	Testing and verification
Week 8	Result analysis, documentation and presentation

Team Roles

Team Member Major Responsibility

S. Jefna Princy Module design, implementation and integration

S. Aswini Data preparation and analytics support

Kavya Dashboard and reporting support

Vishali Testing and verification

Yatra Documentation and presentation

Estimated Budget

Resource	Estimated Cost
Existing Computer/Laptop	₹0
Open-source Software/Tools	₹0
Dataset	₹0
Internet/Cloud Usage	[Actual Cost]
Documentation/Printing	[Actual Cost]
Total	[Insert Actual Total]

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

CanteenCast ZeroWaste – Module 5, Smart Demand Forecasting and Food Waste Management, provides a data-driven approach for improving food preparation and reducing unnecessary food wastage in canteen operations.

The module uses historical food order and sales data to analyze consumption patterns and estimate the expected demand for different food items. The generated demand estimates help canteen staff plan suitable preparation quantities and reduce the possibility of excessive food production or food shortages.

In addition to demand forecasting, the module records and analyzes food waste information. The analysis helps identify frequently wasted food items and recognize wastage patterns. Combining demand forecasting with waste analysis provides a better understanding of food consumption and supports more efficient resource utilization.

The implemented module demonstrates how data analysis and cloud-based system capabilities can be applied to a practical food-management problem. Although prediction accuracy depends on the quality and availability of historical data, the system provides useful decision-support information for canteen management.

Overall, Module 5 contributes to the objectives of CanteenCast ZeroWaste by supporting better demand planning, reducing avoidable food waste, improving resource utilization, and promoting sustainable canteen management.

5.2 Future Work

The following improvements can be considered for future development:

- Improve forecasting accuracy using larger and more diverse historical datasets.
- Apply advanced machine learning models for more accurate demand prediction.
- Include factors such as holidays, special events, weather, and seasonal demand.
- Provide real-time updating of predictions using current order information.
- Develop automated alerts when predicted demand or food waste exceeds a defined level.
- Integrate inventory information with demand forecasting to improve ingredient planning.
- Add more advanced dashboards and visualization features for management.
- Continuously compare predicted demand with actual consumption to improve future predictions.

These improvements can make the system more adaptive, accurate, and useful for large-scale

canteen operations.

5.3 Professional Learning and Individual Reflection

The development of Module 5 provided practical experience in applying engineering concepts to a real-world food-management problem. The work involved understanding requirements, analyzing historical data, designing a forecasting and waste-management workflow, implementing the required functionality, and testing the generated results.

Through this module, practical knowledge was gained in data analysis, demand forecasting, food waste monitoring, system design, testing, and interpretation of analytical results. The development process also demonstrated the importance of data quality because inaccurate or incomplete input data can directly affect the reliability of system outputs.

The project improved problem-solving and engineering decision-making skills by requiring a balance between prediction accuracy, data availability, system simplicity, and practical usability. It also provided experience in understanding how software-based solutions can contribute to sustainability by reducing unnecessary resource consumption and food wastage.

The overall experience strengthened technical, analytical, documentation, testing, and teamwork skills and provided practical understanding of developing a data-driven engineering solution for a real-world application.

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APPENDICES

Appendix A – Sample Historical Order Data

A sample dataset containing historical food orders and sales information used for demand analysis and forecasting.

Date	Food Item	Quantity Ordered	Quantity Sold
Day 1	Meals	100	92
Day 2	Meals	110	105
Day 3	Meals	95	90
Day 4	Meals	120	112
Day 5	Meals	105	98

Appendix B – Demand Forecasting Output

The output generated by the demand forecasting component showing the estimated quantity required for selected food items.

Paste the screenshot of the actual demand forecasting output here.

Appendix C – Food Waste Records

Sample records used to monitor and analyze food wastage.

Date	Food Item	Prepared Quantity	Waste Quantity
Day 1	Meals	100	8
Day 2	Meals	110	5
Day 3	Meals	95	5
Day 4	Meals	120	8
Day 5	Meals	105	7

Appendix D – Food Waste Analysis Output

The output generated from the food waste analysis showing wastage patterns and frequently wasted food items.

Appendix E

Screenshots showing the implementation and working of the Smart Demand Forecasting and Food Waste Management module.

Appendix F – Testing Evidence

Screenshots or outputs showing the testing and verification of demand forecasting, food waste recording, analysis, and report generation functions.